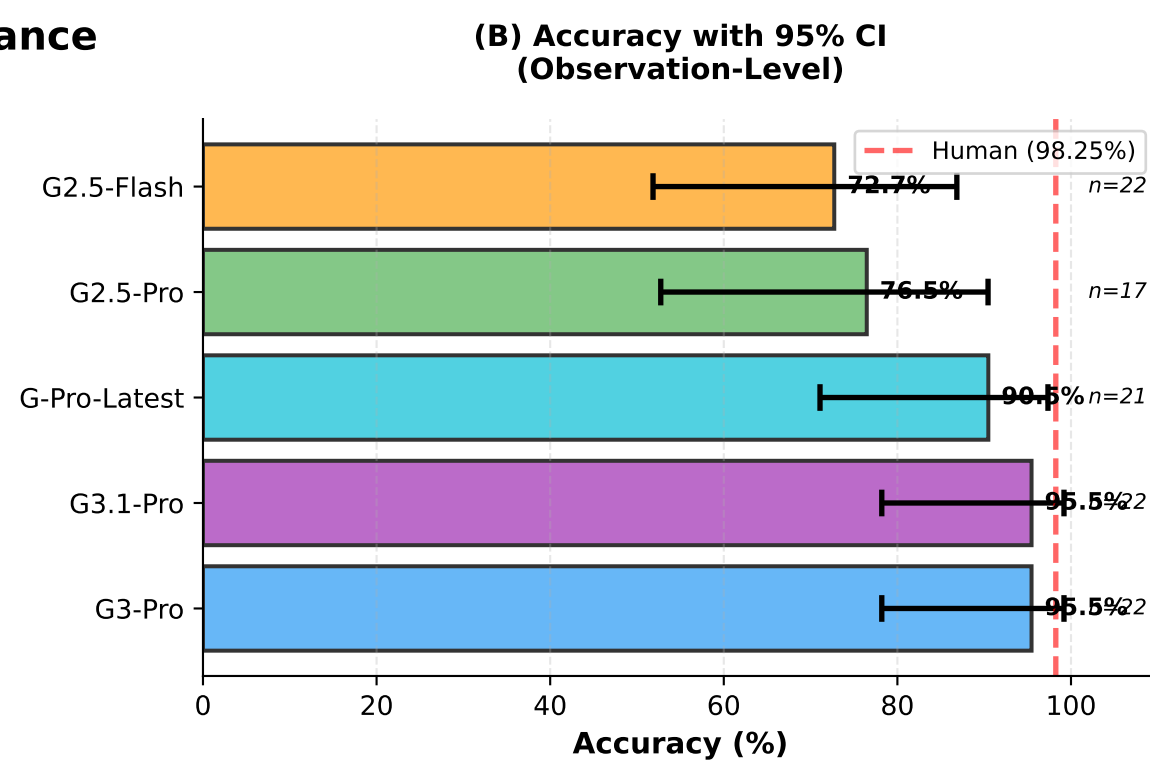
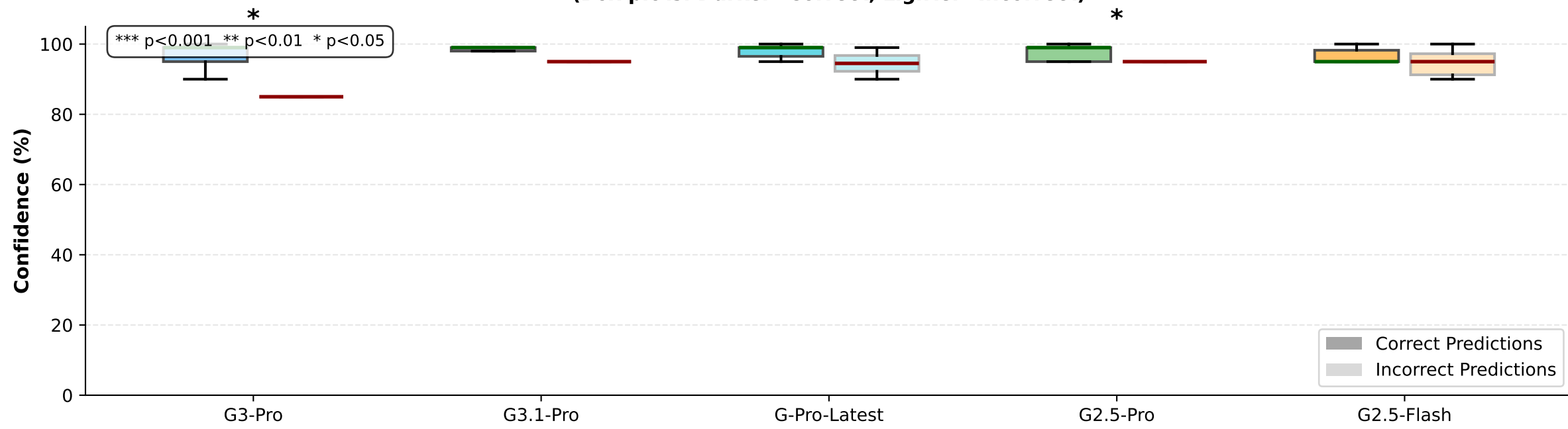


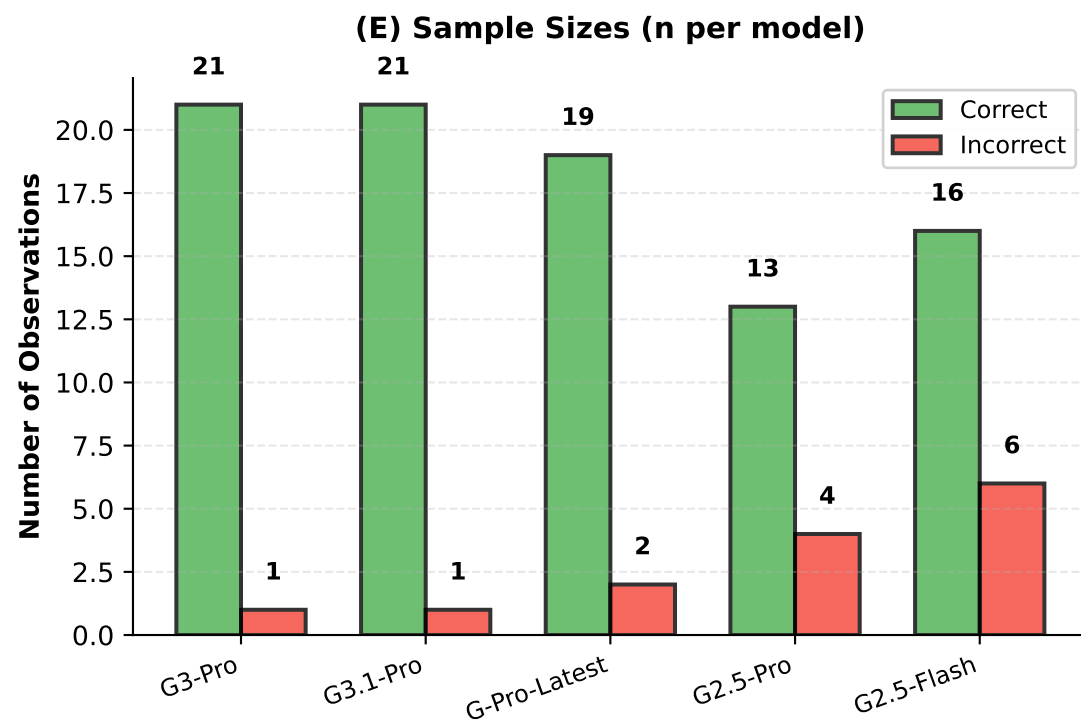
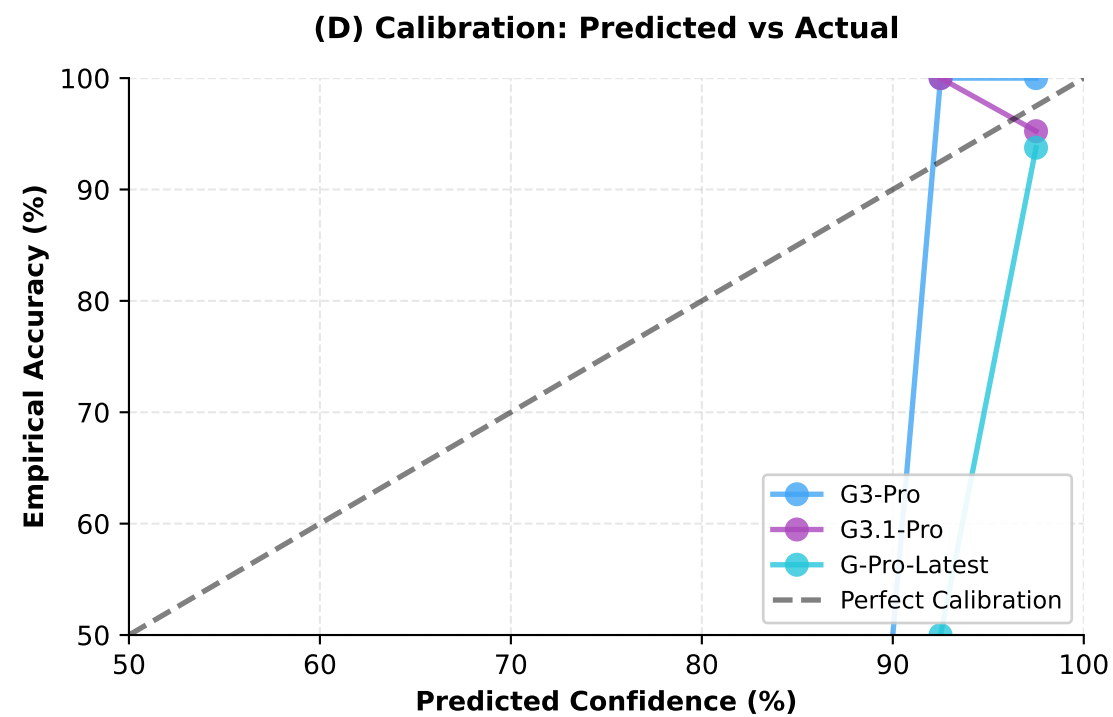
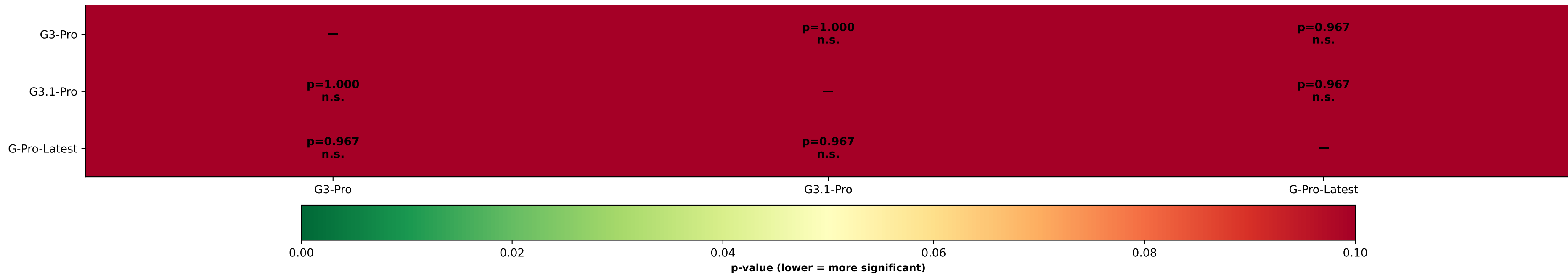
Observation-Level Statistical Analysis: Multi-Model VLM Performance

(Box plots: Darker=Correct, Lighter=Incorrect)



(C) Pairwise Model Comparison (Chi-Square Test)

Green=Significantly Different, Red=Not Significant



STATISTICAL SUMMARY

=====

Total Observations: 104
Models Evaluated: 5

BEST MODEL: G3-Pro

- Accuracy: 95.45%
- n = 22 observations
- Correct: 21
- Incorrect: 1

CONFIDENCE ANALYSIS (All Models Combined):

- Median (Correct): 99.0%
- Median (Incorrect): 95.0%

MANN-WHITNEY U TEST:

- U statistic: 953
- p-value: 6.09e-04

INTERPRETATION:
p=6.09e-04 means
0.061% chance this
pattern occurred by
random luck.

✓ Confident predictions
ARE significantly more
likely to be CORRECT.

STATISTICAL POWER:
✓ HIGH (n=104 total
observations, not n=8)